

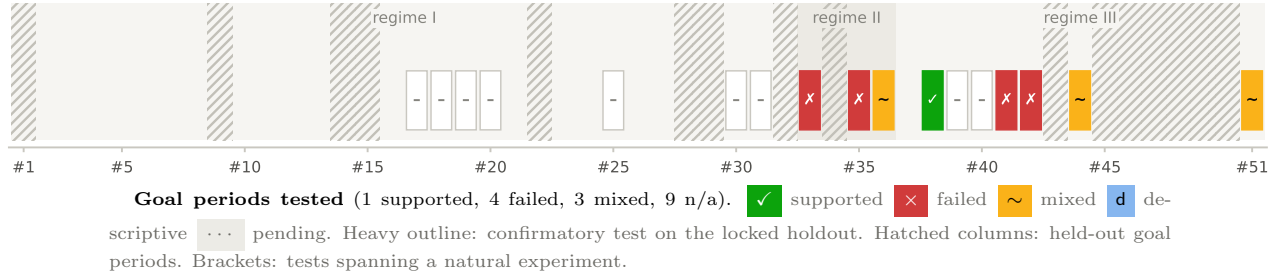
H33: The diversity–productivity curve is an inverted U: the swarm’s operating point

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 0.5/5 Usefulness 1.5/5

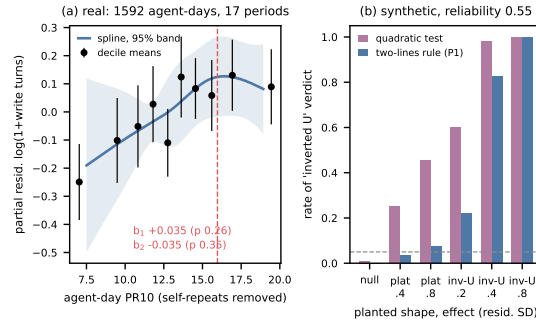
Bottom line. Agent-day content diversity has no output optimum in write turns or landed commits; curvature hurts out of sample, and the one right-side drop is a #51 property.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H33 asks whether an agent’s daily content diversity has an output optimum. The unit is the agent-day. Its diversity is the participation ratio PR10 of its chat: self near-copies removed, rarefied to 10 statements, whitened bge. Its output is $\log(1 + \text{write turns})$. The fit is within agent and period, with day effects and activity controls. The data are 17 non-holdout periods with logged write verbs and 1592 agent-days. The primary test is Simonsohn’s two-lines test. A validation on the real design gives 0–7% false positives and 82% power for a 0.4-SD inverted U. **The data show no inverted U.** The slope is $+0.035$ ($p = 0.26$) below the breakpoint and -0.035 ($p = 0.35$) above it. The curve is flat from PR10 ≈ 14 up. Per period: 0 supported, 3 mixed, 11 failed; the meta-analysed right slope has the wrong sign. PR10 improves day-blocked CV by 0.2%, and curvature makes it worse. Self-repetition does not predict same-day output ($p = 0.84$). **Round 1b** (work commits, #30+, both embedding models) keeps the null. The holdout script is frozen, not run.



Observables. (a) Partial residuals of $\log(1 + \text{write turns})$ by PR10 decile, with the 4-df spline. Agent×period effects, day effects and activity controls are removed. The curve rises a little (not significant) and is flat above PR10 ≈ 14 . Dashed line: two-lines breakpoint. (b) Synthetic replay on the real design at PR10’s measured reliability (0.55). The two-lines rule rarely calls a plateau an inverted U; the quadratic test does so 25–46% of the time. The rule detects a 0.4-SD inverted U 82% of the time.

PHYSICAL MODEL

Model 11 (vector spins). The degree of freedom is a chat statement: a unit vector in the regime’s whitened 32-d bge space. The covariance Σ of an agent-day gives the participation ratio PR, the control parameter. PR = 1 is a frozen spin that repeats itself; PR = 32 is isotropic (“paramagnetic”) content. H33 claims that output peaks at an intermediate PR. The output is $y_{id} = \log(1 + \text{writes})$, with writes = git push or commit, deploy, PR/MR create or merge. α_{iu} and δ_d are agent×period and day effects; z holds activity controls. x_c is the Robin Hood breakpoint; $\nu = n - 1$, $n = 10$.

$$\text{PR} = (\text{tr } \Sigma)^2 / \text{tr } \Sigma^2 \quad (\hat{a}, \hat{b} \text{ unbiased from } W = \nu S)$$

$$y_{id} = \alpha_{iu} + \delta_d + f(\text{PR}_{id}) + \gamma^\top z_{id} + \varepsilon_{id}$$

$$f = b_1(x - x_c)[x < x_c] + b_2(x - x_c)[x \geq x_c] + c[x \geq x_c]$$

inverted U: $b_1 > 0$, $b_2 < 0$ ($p < .05$), $\arg \max f$ interior

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

No content-diversity operating point exists to steer toward. Do not expect more or fewer chat topics to change how much agents ship. Write turns at the best PR10 and at the 90th percentile differ by at most about 9%. The only hint is on the loop side. The bottom decile may write about 25% less (not significant). H12’s self-repetition rate catches loops more directly. Daily chat PR is half noise, and write turns count activity, not value. This null removes a plausible dashboard metric.

RESULTS VS PREDICTIONS

Prediction	Round 1 (write turns) → 1b Verdict (work commits)
P1 inverted U	$b_2 -0.035$ ($p .35$) → -0.040 ($p .31$); $\times$ b_1 n.s.
P2 per period	meta $b_2 +0.028$ → $+0.017$ (wrong $\times$ sign); 1/8 supported
P3 curvature in CV	spline worse than linear (5/5 seeds) $\times$ both rounds
P6 loops cost out-put	$+0.17$ → $+0.97$ (n.s.); $-$ in #31, $\times$ #35, #42
P7 swarm-day	$b_1 -0.22$ → -0.25 $\times$
gte / dedups (new)	$b_2 < 0$ in all 8 variants; sig. in 3 bge #51 specs, none gte; 0 without #51 tail
G51 rival pairs (native)	same role, same day: quad $p = 0.58$ null tive)
G51 role classes (native)	support class (4 agents) concave, $p < .001$ lead
G39 own worlds (native)	$\rho(\text{PR10, commits}) +0.10$ null
G42 videos (native)	$\rho(\text{PR10, videos}) +0.10$ null

GOAL PERIODS IN DETAIL

G33 $\times$ 1b: failed (work commits: $b_1 -0.076$, $b_2 +0.206$ (p 0.3...
G35 $\times$ 1b: failed (work commits: $b_1 +0.019$, $b_2 +0.068$ (p 0.9...
G36 $\sim$ 1b: mixed (work commits: $b_1 +0.154$, $b_2 -0.132$ (p 0.36...
G38 $\checkmark$ 1b: supported (work commits: $b_1 +0.050$, $b_2 -0.166$ (p ...
G41 $\times$ 1b: failed (work commits: $b_1 +0.007$, $b_2 +0.508$ (p 0.0...
G42 $\times$ 1b: failed (work commits: $b_1 -0.044$, $b_2 +0.425$ (p 0.1...
G44 $\sim$ 1b: mixed (work commits: $b_1 +0.087$, $b_2 -0.046$ (p 0.54...
G51 $\sim$ 1b: mixed (work commits: $b_1 +0.037$, $b_2 -0.061$ (p 0.32...
G17 $-$ 1b: n/a (pre-#30: work-ledger zeros ambiguous; 2026-1...
G18 $-$ 1b: n/a (pre-#30: work-ledger zeros ambiguous; 2026-1...

+7 more periods: see the card.

FAITHFULNESS SCORECARD

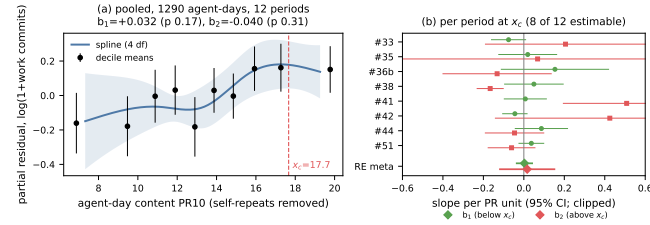
A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The work ledger is dense only from #30. Five round-1 periods drop out, and #51 is 58% of the pool. Agent-day PR10 depends moderately on the embedding model (bge vs gte $\rho = 0.77$) and is half noise (reliability 0.55). The right-side drop appears only in some bge specifications and vanishes without #51. The support-class inverted U rests on 4 agents. Commits count landed work, not value.

ROUND 2



Second figure. Round 1b. (a) Partial residuals of $\log(1 + \text{work commits})$ by PR10 decile, with the 4-df spline and the two-lines breakpoint. The pool is 12 eligible periods from #30 on, 1,290 agent-days. (b) Per-period slopes below and above the breakpoint, with the random-effects meta. The meta right slope has the wrong sign.

Keep this null closed unless the holdout confirms the #51 tail lead (C5). If any test runs, test the support-role curve on the #51 tail. The only untried angles are hour-scale designs and Jev progress scores.

No round-2 redirects yet.

H33-diversity-productivity · generated 2026-10-04 12:00 by infra/summaries/build_summaries.py · card hypotheses/H33-diversity-productivity/README.md · ratings: H33 round-1b agent (suggested) (page 2)